Q1 — Dataset Selection

1. Dataset Name

Global E-Commerce Sales & Customer Behavior Dataset

2. Source and URL

Source: Kaggle.com
Dataset URL: https://www.kaggle.com/datasets/hanzalamushtaq/50k-global-retail-sales-and-customer-ratings-data?resource=download

3. Number of Rows and Columns

- **Rows:** 50,000
- **Columns:** 15

4. Brief Description of the Dataset

his dataset contains 50,000 highly realistic, synthetic e-commerce transaction records spanning from January 2023 to June 2026. It captures detailed customer demographics, multi-category purchasing behavior, pricing dynamics, seasonal sale spikes, and shipping performance indicators. It is perfectly tailored for Exploratory Data Analysis (EDA), customer segmentation, and predictive machine learning tasks.

5. Why I Selected This Dataset

I selected this dataset because it contains 50,000 transaction records and provides sufficient detail to investigate sales performance, customer purchasing behaviour, product demand, and geographic patterns. Its time-based and transactional structure supports meaningful analysis, while its data-quality issues provide opportunities to demonstrate practical data-cleaning and preprocessing skills.

6. Business Opportunities or Problems to Investigate

The dataset can help investigate the following business opportunities and problems:

1. **Sales trends:** Identify monthly and seasonal changes in sales revenue and transaction volume.
2. **Product performance:** Determine which products contribute most to revenue and which products have comparatively low sales.
3. **Customer behaviour:** Identify repeat customers, high-value customers, and differences in purchasing patterns.
4. **Geographic performance:** Compare sales across countries to identify potential markets for business expansion.
5. **Returns and cancellations:** Examine cancelled transactions and their potential impact on reported sales.
6. **Revenue concentration:** Investigate whether a small number of products or customers account for a disproportionate share of revenue.